Program : **Diploma in Cyber Forensics and Information Security**	
Course Code : **6288**	Course Title: **Advanced Digital Forensics Lab**
Semester: **6**	Credits: **2.5**
Course Category: **Programme core course**	
Periods per week: **4 (L: 1 T: 0 P: 3)**	Periods per semester: **60**

Course Objectives:

- To explain the need for software security
- To explain the various types of security attacks and the risks associated.

Course Prerequisites:

Topic	Course Code	Course Name	Semester
Knowledge about various forensics tools	4289	Cyber Forensics Lab	4

Course Outcomes:

On completion of the course student will be able to:

COn	Description	Duration (Hours)	Cognitive Level
CO1	Develop in depth knowledge of Computer Forensics	13	Applying
CO2	Develop in depth knowledge of Computer Forensics tools	13	Applying
CO3	Identify digital evidences	15	Applying
CO4	Analysis and validating of forensic data	16	Applying
	Lab Test	3	

CO – PO Mapping

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	2	2	2			1	
CO2	2	2	2			1	
CO3	2	2	2			1	
CO4	2	2	2			1	

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline

Module Outcomes	Name of Experiment	Duration (Hours)	Cognitive Level
CO1	**Develop in depth knowledge of Computer Forensics**		
M1.01	Use a Web search engine, such as Google or Yahoo!, and search for companies specializing in computer forensics. Select three and write a two- to three-page paper comparing what each company does	1L+12P	Understanding Applying
CO2	**Develop in depth knowledge of Computer Forensics tools**		
M2.01	Search the Internet for articles on computer crime prosecutions. Find at least two. Write one to two pages summarizing the two articles and identify key features of the decisions you find in your search	2L+4P	Understanding Applying
M2.02	Use a Web search engine, search for various computer forensics tools.	2L+5P	Understanding Applying
	Lab Test – I	1.5	
CO3	**Identify digital evidences**		
M3.01	Preparing and processing of investigations. Try to examine and identify the evidences from the drives	1L+4P	Understanding Applying

M3.02	Extracting of files that have been deleted	1L+9P	Understanding Applying
CO4	**Analysis and validating forensic data**		
M4.01	Illustrate the analysis of forensic data.	1L+4P	Understanding Applying
M4.02	Illustrate the validating of forensic data	1L+4P	Understanding Applying
M4.03	**-Open ended experiments	6	
	Lab Test – II	1.5	

** - Sample Open Ended Experiments (Not for End Semester Examination but compulsory to be included in Continuous Internal Evaluation. Students can do open ended experiments as a group of 2-3. There is no duplication in experiments between groups. Open ended experiments should include the examination and identification of digital evidences

1) Extracting browsing artifacts from web browser log files.

Text / Reference

T/R	Book Title/Author
T1	Robert C. Seacord, "Secure Coding in C and C++ (SEI Series in Software Engineering)",Addison-Wesley Professional, 2005.
T2	Jon Erickson , "Hacking: The Art of Exploitation", 2nd Edition, No Starch Press, 2008.
T3	Mike Shema, "Hacking Web Apps: Detecting and Preventing Web Application SecurityProblems", First edition, Syngress Publishing, 2012

Online Resources

Sl.No	Website Link
1	https://onlinecourses.swayam2.ac.in/aic20_sp06/preview
2	https://onlinecourses.swayam2.ac.in/arp19_ap79/preview